

Semiconductors - Asia-Pacific

OCP APAC Summit: Agentic AI, CPO and advanced packaging integration in focus

Industry Overview

From chip to grid: industry leaders chart the next AI era

We attended 2026 OCP (Open Compute Project) APAC Summit on Aug 11-12. The event emphasizes the limiting factor of AI innovation is across the silicon supply constraint to the capability of the system level integration across the supply chain. The speakers (e.g. Microsoft, Google, TSMC, ASE, GlobalFoundries, ARM, Nvidia, AMD, Broadcom, GUC, Arista, Ayar Labs) expect faster development in CPU, ASIC, optical interconnect/CPO through front-end manufacturing, packaging/testing and power system level integration.

Agentic AI supporting ASICs/CPUs outgrowing GPUs

As AI CPU usage intensity grows, the industry expects the configuration of GPU/ASIC vs. CPU to evolve from 4:1 to 2:1 and 1:1 in the future. Microsoft highlighted its Cobalt 200 CPU with 33% lower agent-call latency, 50% more performance vs. Cobalt 100 and 23% higher agentic throughput while its Maia 200 ASIC adopted in GPT5.5 and 365 Copilot delivers 30% better token/dollar vs. GPUs. ARM sees chiplets as alternative to SoCs requiring US\$1bn+ design cost and 36 months design cycle and the CPUs built on its Neoverse V3 cores and total design platform support faster hyperscalers' deployment.

CPO closer to deployment; pluggable remain competitive

The speakers see the optics are enabler for compute and HBM system utilization and power efficiency and expect fast upgrades from 800G to 1.6T and CPO. Nvidia's CPO infrastructure is in production while Broadcom also sees CPO providing 10x reliability improvement and 70% power reduction vs. pluggable. However, Ayar Labs pointed out that the optical engine could be placed on-board, on-package and on-interposer, suggesting progressive migration rather than a binary pluggable to CPO transition.

Packaging & testing content rises as migration accelerates

AI packaging is moving toward larger panels, 3D/hybrid bonding, CPO and integrated power. ASE sees the panel vs. wafer improve large reticle size output meaningfully and vertical power delivery to reduce power loss from 10% to 2% while 3D face to face hybrid bonding can deliver ~10x higher die to die interconnects density, allowing stacked logic dies to behave like single integrated SoC. The packaging upgrades also come with a cost of yield loss in the late stage, making additional wafer/die/package/system level test required for the systems integrating multiple chipsets, HBM and optics components.

Stocks: prefer TSMC, ASE, Chroma, Mediatek, GUC

OCP takeaways consistent with the supply chain aggressive capex on strong demand and technology upgrade. We keep preference on the companies with high CPU/ASIC and CPO exposure potentially outgrowing in the AI space. Next catalyst: Semicon Taiwan in Sept.

13 August 2026

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Refer to important disclosures on page 7 to 11. Analyst Certification on page 4. Price Objective Basis/Risk on page 3.

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Timestamp: 12 August 2026 06:00PM EDT

Exhibit 1: Terms and definitions

Glossary of abbreviations/acronyms

Term	Definition
AI	Artificial Intelligence
AMD	Advanced Micro Devices
AOI	Automated Optical Inspection
API	Application Programming Interface
ASIC	Application Specific Integrated Circuit
ATC	Active Thermal Control
CoPoS	Chip-on-Panel-on-Substrate
CoWoS	Chip-on-Wafer-on-Substrate
CoWoS-L	Chip-on-Wafer-on-Substrate-Local Silicon Bridge
CPO	Co-Packaged Optics
CPU	Central Processing Unit
DC	Datacenter
EFB	Enhanced Fan-out Bridge
FoCoS	Fan-Out Chip on Substrate
FOPoP	Fan-Out Package on Package
FOSiP	Fan-Out System-in-Package
FT	Final test
GPU	Graphic Processing Unit
HBM	High Bandwidth Memory
HPC	High-Performance Computing
I/O	Input/Output
IaaS	Infrastructure as a Service
IC ATM	IC Assembly, Testing, and Material
IDM	Integrated Device Manufacturer
LEAP	Leading-Edge Advanced Packaging
MCL	Micro channel lid
MPU	Microprocessor Unit
NB	NoteBook
OSAT	Outsourced Semiconductor Assembly/Testing
PC	Personal Computer
RDL	Redistribution Layer
SLT	System level test
SoC	System on Chip
SoIC	System on Integrated Chips
SoIC-MH	System-on-Integrated-Chips-Molding-Horizontal
SWIFT	Silicon Wafer Integrated Fan-out Technology
TAM	Total Addressable Market
TPU	Tensor Processing Unit
WMCM	Wafer-level-multi-chip-module
WPM	Wafer per month
WPY	Wafer per year
XPU	Accelerating Processing Unit

Source: BofA Global Research estimates

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Stocks mentioned

Prices and ratings for stocks mentioned in this report

BofA Ticker	Bloomberg ticker	Company name	Price	Rating
XSRIF	3711 TT	ASE Technology	NT\$ 621	C-1-7
CRMJF	2360 TT	Chroma ATE	NT\$ 2195	C-1-7
GBUHF	3443 TT	Global Unichip Corp.	NT\$ 5130	C-1-7
MDTKF	2454 TT	MediaTek	NT\$ 4015	C-1-7
TSMWF	2330 TT	Taiwan Semi Mfg Co	NT\$ 2415	B-1-7

Source: BofA Global Research

Price objective basis & risk

ASE Technology Holding (XSRIF)

We value ASE Technology Holding at NT\$750 per share (US\$48 for ADR), based on 20x 2H27-1H28E P/E, above its historical range of 5-26x, in view of its solidified industry position in AI era. We use P/E ratio to value ASE as the company has been able to maintain healthy profitability through the cycles in the past 20 years due to consolidating industry landscape supporting a more benign supply/demand and keeping pricing and margins less cyclical.

Downside risks to our PO are 1) share loss and/or ASP erosion owing to Chinese players' cannibalization and/or ASE's worsening execution, 2) weakening end demand due to uncontrollable matters, such as macro halt and/or geopolitical tensions, 3) regulatory issues that could hinder ASE from generating synergies and benefit its competitors.

Chroma ATE (CRMJF)

Our price objective of NT\$3,050 on Chroma is based on 38x 2027E P/E, which is at the top of the company's historical trading range of 12-39x. In our view, the valuation multiple is underpinned by

1) +70% earnings CAGR in 2025-27E supported by +75% sales CAGR growth and GMs/OpMs expanding, 2) ROE expansion to around 35% in 2025-27E, 3) net cash of around NT\$21/share, and 4) free cash flow generation of around NT\$15/share per annum across 2025-27E.

Downside risks to our price objective are:

1) weaker market demand, especially the cloud AI investment moderation, leading to slower infrastructure builds especially in China and dragging the company's revenue growth for its power testing equipment business and operating leverage
2) a more conservative investment cycle from the hyperscalers on further data center deployment due to growing capex could dampen the demand for GPU and ASIC, negatively impacting the testing equipment demand for related chipsets.
3) failure to execute on the customized equipment requested by its customers or delay of technology roadmap could lead to an order shift to its competitors.

Global Unichip Corp. (GBUHF)

Our PO NT\$7,000 is based on 50x 2H27-1H28E P/E. This PE multiple is around the high-end of the historical trading range of 10-70x from 2013-2026, on the back of: 1) around 88% earnings CAGR during 2026-28E, 2) 46% ROE during 2026-27E, 3) robust cash levels, 4) operating margin of 15% into 2026-28E.

Upside risks are

(1) stronger than expected revenue from substrate design business
(2) stronger for longer demand from existing mass production projects, e.g. SSD (solid state drive) controller, networking chip

Downside risks:

(1) further delay of NRE (non recurring engineering) projects which could subsequently lead to the delay in entering into turnkey stage (and thus slower operating margin expansion)
(2) sudden shutdown or halt in China-related business given the geopolitical tensions
(3) effects of low-margin cryptocurrency ASIC wind down are worse than expected
(4) Weaker than expected shipment of Google's TPU, leading to lower turnkey revenue contribution
(5) Unexpected delay in ADAS project development, leading to slower ramp in revenue contribution from 2027

MediaTek (MDTKF)



Our PO of NT\$5,230 is based on 23x 2H27-1H28 P/E, as we are constructive on its broadening ASIC pipeline to drive earnings upgrade cycle and re-rating as it locks in more supply.

Upside risks to our PO are: (1) better than expected smartphone demand, especially in China, (2) a better pricing in mobile communication segment thanks to alleviated competition and rising exposure to high-end/flagship market, and (3) an even more superior investment return on non-op side to uplift its net cash position. (4) Unexpected collaboration with global technology leaders leads to higher than expected margins/profits

Downside risks to our PO are: (1) weaker-than-expected end-market situation in consumer-related smartphone/TV space, (2) macro challenges including inflation/policy-related pressures, (3) rising competition from existing players and/or newcomers in smartphone/TV SoC (system on chip) space, and (4) ASP/margin pressure caused by each or all of the above and/or rising foundry costs. (4) Bidding process for Meta Inference ASIC is slower than expected.

Taiwan Semiconductor Manufacturing Co. (TSMWF)

Our PO for TSMC is NT\$3,100 per share (US\$590 per ADR), based on 20x 2027E P/E. TSMC's valuation has traded in the range of 10-26x PE from 2017. 20x is in the mid to high end of the historical range, above the average P/E of 18x. We think a PE above the average is justified given TSMC's enhanced industry position and mid-40% 5-year CAGR in AI should underpin a rerating

Downside risks to our price objective are: (1) greater-than-expected slowdown in global smartphone/consumer electronics demand, (2) Intel's potential insourcing strategy and ambitions in foundry service, (3) execution risks on advanced nodes, and (4) higher than expected tariffs.

Analyst Certification

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Investment rating	Company	BofA Ticker	Bloomberg symbol	Analyst
BUY				
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	All Ring	XNQNF	6187 TT	Haas Liu
	AMEC	XRTOF	688012 CH	Dai Shen
	ASE Technology Holding	XSRIF	3711 TT	Haas Liu
	ASE Technology Holding -ADR	ASX	ASX US	Haas Liu
	ASMPT	ASMFV	522 HK	Simon Woo, CFA
	Aspeed	XLKMF	5274 TT	Mike Yang
	Chroma ATE	CRMJF	2360 TT	Haas Liu
	eMemory	XYLWF	3529 TT	Mike Yang
	GigaDevice	XGXIF	603986 CH	Daley Li, CFA
	Global Unichip Corp.	GBUHF	3443 TT	Haas Liu
	Grand Process Technology Corp	XZOWF	3131 TT	Haas Liu
	Hanmi Semiconductor	HNSIF	042700 KS	Simon Woo, CFA
	Hon Precision	XXPIF	7769 TT	Haas Liu
	Horizon Robotics	HRZRF	9660 HK	Daley Li, CFA
	Iluvatar CoreX	XSIBF	9903 HK	Daley Li, CFA
	Ingenic	XISCF	300223 CH	Dai Shen
	InnoScience Technology	XSCHF	2577 HK	Daley Li, CFA
	JCET Group Co Ltd	XJIEF	600584 CH	Dai Shen
	King Yuan Electronics Corp.	KYUFF	2449 TT	Haas Liu
	Kinik	KIKCF	1560 TT	Haas Liu
	Macronix International	MXICF	2337 TT	Dai Shen
	MediaTek	MDTKF	2454 TT	Haas Liu
	Montage Technology	XRDF	688008 CH	Daley Li, CFA
	MPI Corporation	XMJCF	6223 TT	Mike Yang
	Nanya Technology	NNYAF	2408 TT	Simon Woo, CFA
	NCE Power	XVFFF	605111 CH	Daley Li, CFA
	OmniVision	XXHQF	603501 CH	Dai Shen
	Phison Electronics	PISNF	8299 TT	Simon Woo, CFA
	Powerchip Semiconductor Manufacturing Co	XCHPF	6770 TT	Haas Liu
	Powertech Technology	PWOGF	6239 TT	Simon Woo, CFA
	Rockchip	XRPF	603893 CH	Daley Li, CFA
	Samsung C&T	XCDTF	028260 KS	Simon Woo, CFA
	Samsung Elec -G	SSNHZ	SMSN LI	Simon Woo, CFA
	Samsung Electronics	SSNLF	005930 KS	Simon Woo, CFA
	Samsung Electronics Preferred	SSNNF	005935 KS	Simon Woo, CFA
	Silergy Corp.	SLEGF	6415 TT	Mike Yang
	Silicon Motion	SIMO	SIMO US	Simon Woo, CFA
	SK Hynix	HXSOF	000660 KS	Simon Woo, CFA
	SK Hynix ADR	SKHY	SKHY US	Simon Woo, CFA
	SK Square	SKSQF	402340 KS	Simon Woo, CFA
	Taiwan Semiconductor Manufacturing Co.	TSM	TSM US	Haas Liu
	Taiwan Semiconductor Manufacturing Co.	TSMWF	2330 TT	Haas Liu
	Vanguard International Semiconductor Co	VGILF	5347 TT	Haas Liu
	Winbond Electronics	WBEKF	2344 TT	Dai Shen
	WinWay Technology	XWCLF	6515 TT	Mike Yang
	WT Microelectronics	XZOPF	3036 TT	Mike Yang
NEUTRAL				
	ASMedia Technology Inc.	XZSFF	5269 TT	Mike Yang
	Black Sesame Intl Holding	BSIHF	2533 HK	Daley Li, CFA
	Hangzhou Silan Microelectronics	XDFRF	600460 CH	Daley Li, CFA
	Lion Electronics	XDHF	605358 CH	Dai Shen
	MetaX	XMEZF	688802 CH	Daley Li, CFA
	Novatek	NVKMF	3034 TT	Mike Yang
	Realtek	RLTKF	2379 TT	Mike Yang
	Shenzhen Goodix	XQPLF	603160 CH	Daley Li, CFA
UNDERPERFORM				
	Crystal Clear	XPPTF	300655 CH	Dai Shen
	Faraday	FDYTF	3035 TT	Haas Liu
	GlobalWafers	XWLF	6488 TT	Mike Yang
	Hua Hong Grace	HHUSF	1347 HK	Dai Shen
	LX Semicon	XLXSF	108320 KS	Simon Woo, CFA
	M31 Technology	XMTZF	6643 TT	Mike Yang



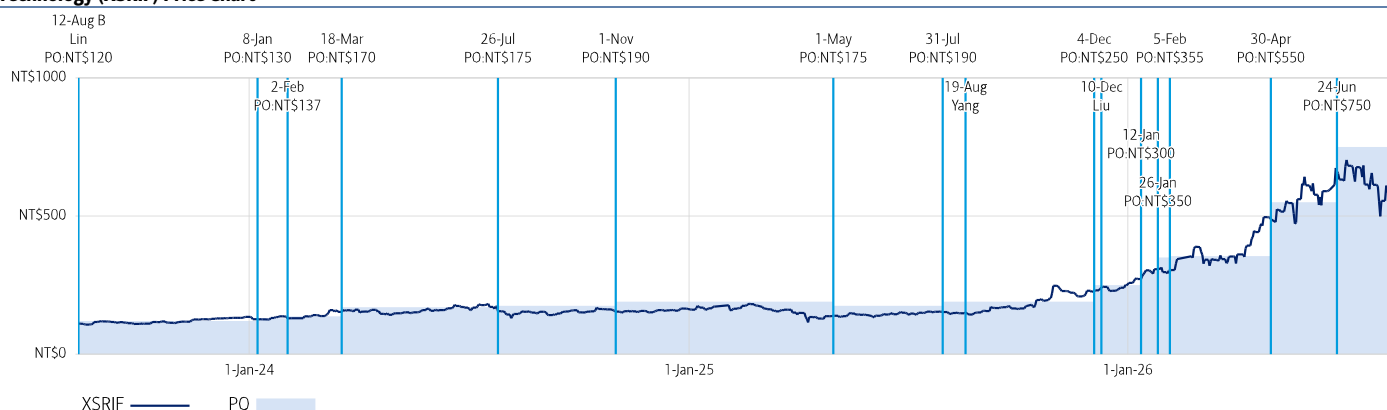
APR - Semiconductor Coverage Cluster

Investment rating	Company	BofA Ticker	Bloomberg symbol	Analyst
	Maxscend	XXSXF	300782 CH	Daley Li, CFA
	Parade	PRDWF	4966 TT	Mike Yang
	Soulbrain	XSBOF	357780 KS	Simon Woo, CFA
	United Microelectronics Corp.	XUMIF	2303 TT	Haas Liu
	United Microelectronics Corp.	UMC	UMC US	Haas Liu
	VeriSilicon	XMLZF	688521 CH	Daley Li, CFA
	Win Semiconductors	WSMIF	3105 TT	Haas Liu
	Wonik IPS	XRHQF	240810 KS	Simon Woo, CFA

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Important Disclosures

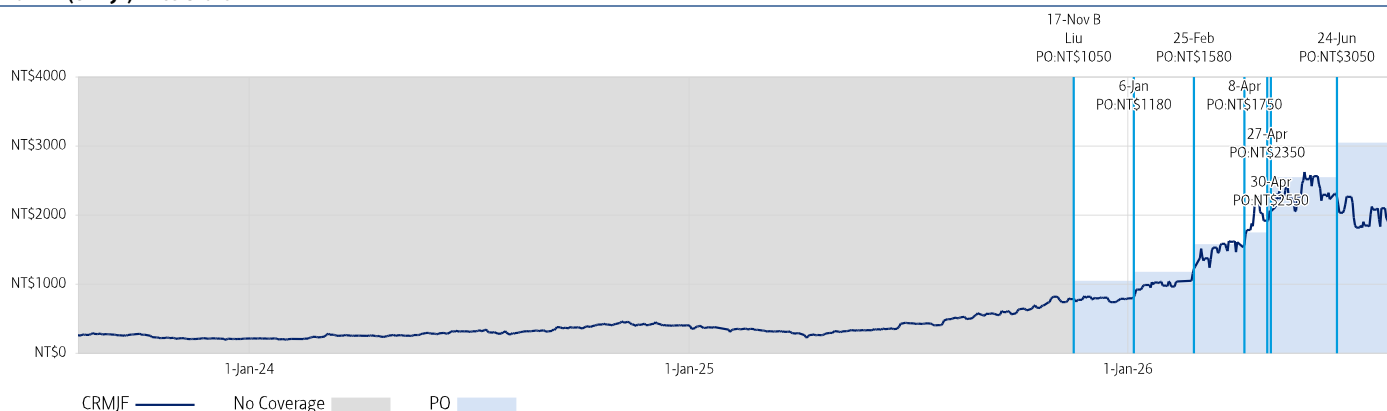
ASE Technology (XSRIF) Price Chart



B: Buy, N: Neutral, U: Underperform, PO: Price Objective, NA: No longer valid, NR: No Rating

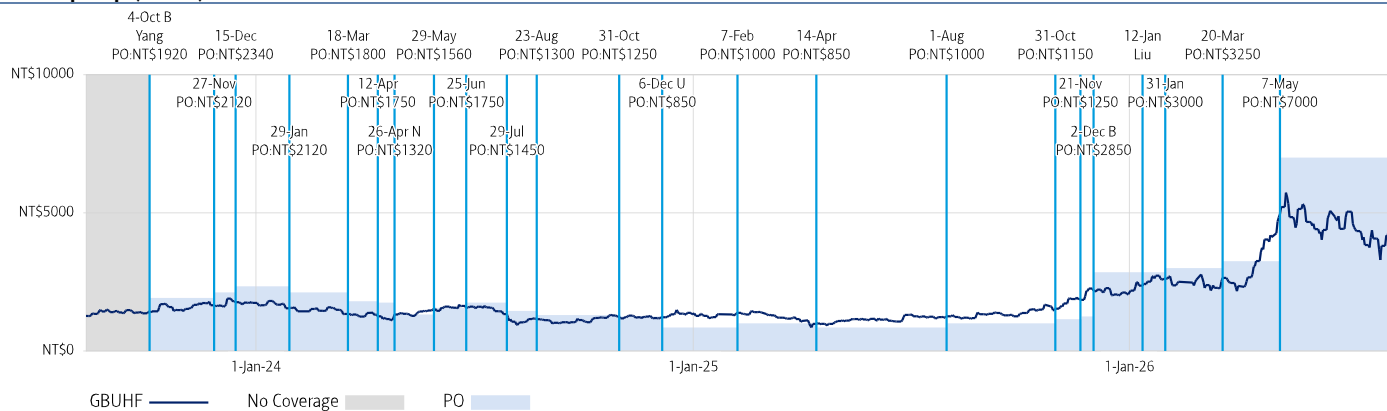
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Chroma ATE (CRMJF) Price Chart

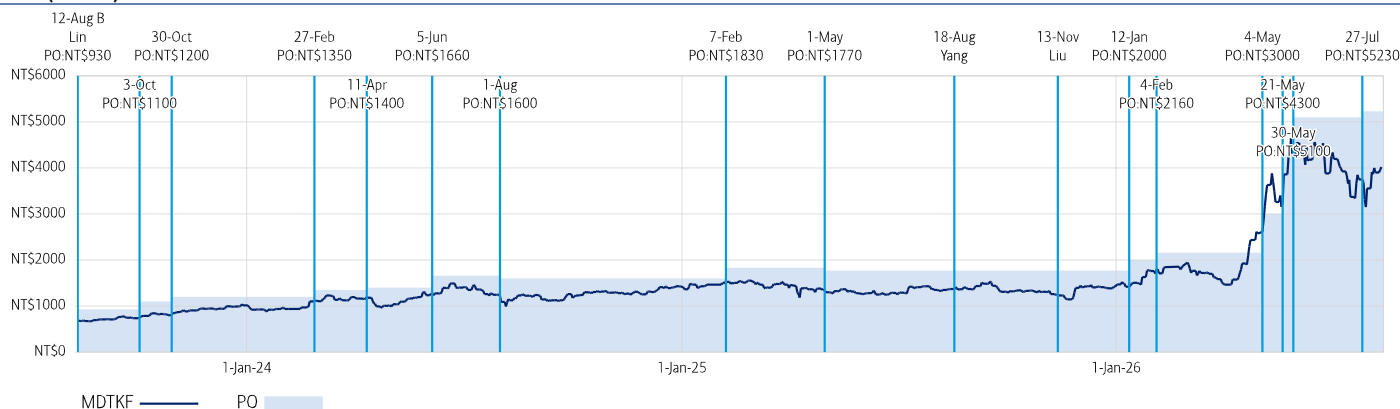


B: Buy, N: Neutral, U: Underperform, PO: Price Objective, NA: No longer valid, NR: No Rating

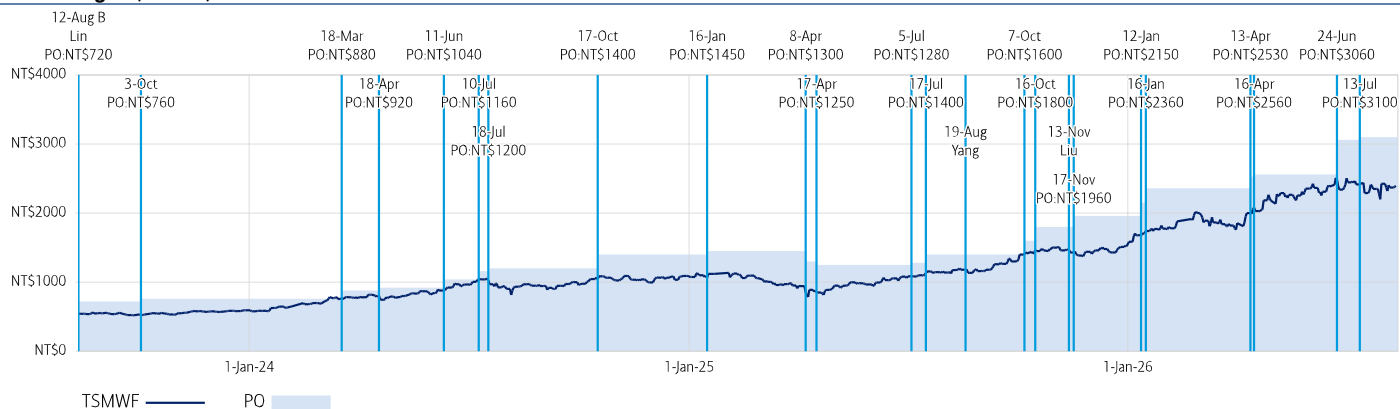
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Global Unichip Corp. (GBUHF) Price Chart

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MediaTek (MTK) Price Chart

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Taiwan Semi Mfg Co (TSMWF) Price Chart

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Coverage Universe	Count	Percent	Inv. Banking Relationships ^{R1}	Count	Percent
Buy	243	60.60%	Buy	123	50.62%
Hold	87	21.70%	Hold	45	51.72%
Sell	71	17.71%	Sell	21	29.58%

Equity Investment Rating Distribution: Global Group (as of 30 Jun 2026)

Coverage Universe	Count	Percent	Inv. Banking Relationships ^{R1}	Count	Percent
Buy	1987	56.11%	Buy	1190	59.89%
Hold	797	22.51%	Hold	496	62.23%
Sell	757	21.38%	Sell	391	51.65%

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Underperform	N/A	≥ 20%

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